

Section 2: System Design Fundamentals

Networking & Communications

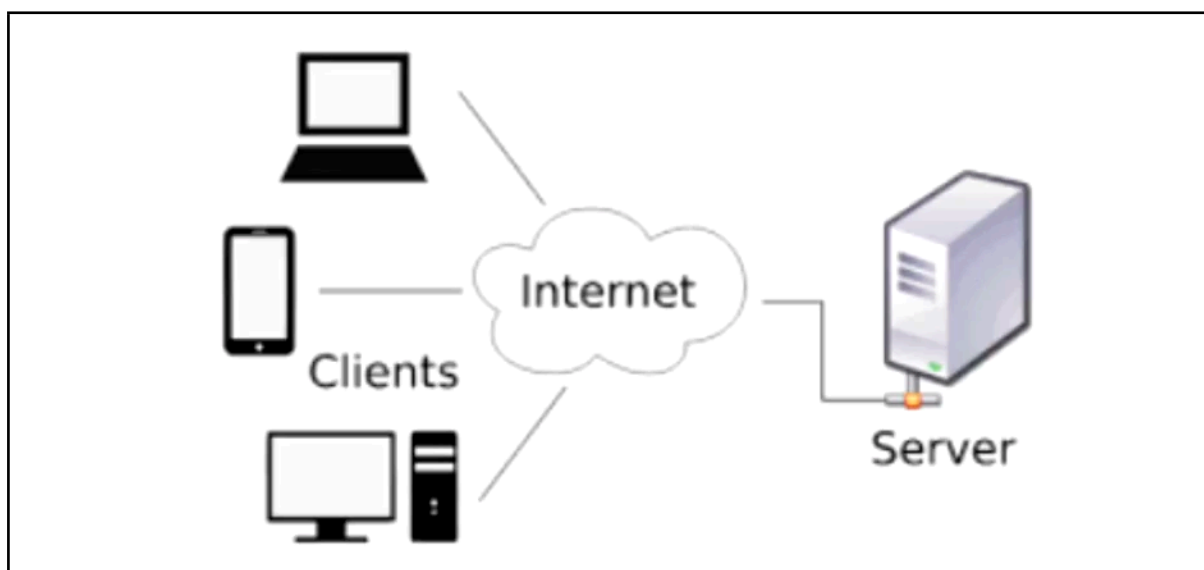
Networking and communications are at the core of every large scale system, whether you are designing a web application, a distributed system, or cloud-based service, your system components must efficiently communicate with each other. Without a strong grasp on networking fundamentals, it is impossible to build scalable, reliable, and high-performance systems.

Throughout this section, we will break down fundamental networking concepts like IP addresses, DNS, client-server communication, proxies, load balancing, API gateway, content delivery networks, and all these concepts are essential for both real world system design problems as well as technical interviews.

By the end of this section, you will have a solid understanding of how networking plays a crucial role in system design, enabling you to make informed architectural decisions and confidently tackle the system design interviews.

Why Networking Matters in System Design?

Imagine using an application like Netflix, WhatsApp, or Amazon. Every time you are streaming a movie on Netflix, you are sending a message in WhatsApp or you are placing an order on Amazon. Millions of data packets are travelling across the network to make that happen seamlessly. This is why networking is a critical component of system design.



Every application is made up of multiple components, and at its core, every system relies on data exchange between these components, whether it's a request from a mobile app to a backend server or communication between distributed microservices.

- Networking ensures that the data flows efficiently.
- A well-designed network infrastructure allows the system to scale to handle millions of users., stays reliable under heavy traffic, and maintains high performance and low latency even with the high traffic.
- Without effective networking, even the best designed-applications can suffer from downtime and slow response times.

There are 4 key areas in system design where the networking plays a crucial role:

- **Communication:** Ensuring smooth data transfer between clients and servers and application servers and databases is the critical aspect of networking.
- **Load Balancing:** Nowadays, we have multiple servers hosting our applications on the internet. Distributing traffic efficiently to prevent overload on a single server is all about load balancing.
- **Security:** We need to protect our data from unauthorized access and cyber threats, and that's where security-related networking protocols and concepts are utilized.
- **Efficiency:** Optimizing network performances to reduce latency and improve user experience is the fundamental point of efficiency in the network design that we do.

Understanding the above concepts, will help us in designing systems that are robust, scalable, and secure.

How Networking Impacts Large-Scale Systems

Have you ever wondered how companies like Netflix, Amazon, or Google handle millions of users at the same time without crashing. The answer lies in efficient networking.

- At scale, a system must serve millions, sometimes billions of users in real time. Networking ensures that the requests are distributed efficiently across servers, preventing bottlenecks and system failures.
- Without robust networking, even the most powerful application would struggle under the high traffic.
- In a large scale system, data is constantly flowing, whether it is retrieving product details from the database, sending messages in a chat application, or loading a web page.
- Optimized networking ensures minimal delays and keeps the user experience smooth and seamless.
- Latency is the time it takes for data to travel between systems. The lower the latency, the better the performance.

- Networking strategies like CDNs, caching, and load balancing help reduce delays and improve resilience, ensuring systems stay operational even during failures.
- Modern applications run on cloud-based and distributed architecture, whether it's microservices communicating over APIs or global users accessing data from a different region. Networking will ensure that everything works together efficiently across different data centers and cloud providers.

Agenda: Key Networking Concepts in System Designs

Now, before we dive into any specific topic, let's take a look at what we'll be covering in this section. Networking is a vast topic, but we will be focusing on the most essential concepts that every system designer must understand.

- Understanding IP addresses
- How DNS Works
- Client-Server Model Explained
- Forward Proxy Vs. Reverse Proxy
- Introduction to Load balancing
- What is an API Gateway?
- Content Delivery Networks (CDN) in System Design